

VitalGuard

Explain the signal. Question the evidence. Keep humans in control.

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THE PROBLEM

Physiological changes and unreliable sensors can look similar. This prototype explores how a monitoring system can explain a change, check the quality of its evidence, and request accountable human review without treating a score as a diagnosis.

THE WORKING PROTOTYPE

Four simulated signals - heart rate, oxygen saturation, respiratory rate and activity - feed a quality-aware pipeline. A per-session median/MAD baseline supports directional anomaly detection. Explicit rules require corroboration and sustained evidence before requesting review. Observation status and review status remain separate.

TRY THE COMPLETE WORKFLOW

Compare activity recovery with a sensor disconnect, then select **Try human review**. Inspect signal contributions and review gates, enter a reason, and confirm or dismiss. Export a JSON session with preserved event evidence and a SHA-256 integrity digest. No external alert or medical action is sent.

EVIDENCE, INCLUDING FAILURE

The reproducible development evaluation contains **644 synthetic traces**. In the noise/baseline-offset cohort, 100/100 anomalies reached review and 0/100 negative traces did; median delay was 40 seconds. **Both failure cohorts missed 40/40 anomalies**: continuous loss produced NO DATA, while jointly misleading oxygen and activity caused silent misses with a high evidence index. These are development results, not clinical accuracy.

IMPLEMENTATION & BOUNDARIES

Browser-native JavaScript shares its deterministic engine with Node.js tests and evaluation scripts. No external model API is required. The evidence index is a heuristic, not calibrated probability. No patient data, device integration, persistent multi-user state, authenticated audit storage or clinical validation is implemented. Next step: validate on appropriate real signals and independent labels.